

Strategic Under-Specification: A Stackelberg Game Between User and Assistant

Pranjal Agarwal
BITS Pilani, Hyderabad Campus

Abstract

Users of coding assistants routinely under-specify their intent, and the assistant’s policy for handling missing information (whether guessing silently or asking a clarifying question) is an endogenous strategic choice that users respond to. We develop a contract-theoretic model of endogenous clarification in human–AI interaction: strategic under-specification is not driven by models becoming better at guessing; it emerges because users strategically economize on specification effort in response to the assistant’s asking policy when the assistant’s objective diverges from true user welfare. Formalizing this interaction as a Stackelberg game between an assistant (leader) and a population of users with privately known specification costs (followers), we first establish a decision-theoretic benchmark with an exogenous policy. By the envelope theorem, improving raw guessing accuracy alone weakly improves user welfare despite shortening prompts, showing that informal claims that better guessing degrades outcomes conflate technological improvement with policy misalignment.

Closing the loop, we solve for the assistant’s optimal policy under two governance regimes: a *pooling* policy restricted to a single population-wide ask rate, and a *screening* policy offering an incentive-compatible menu conditioned on prompt specification effort. We show that an unbiased (welfare-aligned) assistant achieves first-best efficiency under screening despite users’ private cost information, as incentive compatibility is costless when principal and agent share an objective. Consequently, all screening-regime distortion is attributable to misalignment between the assistant’s objective and true user welfare, not to asymmetric information per se. Under pooling on the physically feasible domain ($m \leq k$), an unbiased assistant sets the full asking rate, whereas perceived asking friction suppresses clarification turns ($\partial a^{SE}/\partial \lambda_A \leq 0$). When the assistant is biased, screening induces downward distortion in the asking service offered to high-specification-cost users to prevent low-cost users from mimicking. We characterize both equilibria under a tractable linear-risk specification, evaluate robustness under the exact nonlinear conjunctive specification, and outline a proposed human-subject experimental design.

Keywords: Human–AI Interaction, Contract Theory, Clarifying Questions, Stackelberg Games, Adverse Selection, Moral Hazard.

Code Availability: The simulation suite, symbolic proofs, and mathematical verification scripts are openly available at <https://github.com/pranjal-2706/paper>.

AI-Use Disclosure: Generative AI tools were used for code testing, script execution, and syntax verification. All conceptual formulations, economic arguments, and mathematical proofs were developed and verified by the author.

1 Introduction

A user typing a coding request faces a fundamental economic trade-off: spend cognitive effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to

guess or to ask. Neither choice is made in a vacuum: the user’s specification effort depends on what they expect the assistant to do with silence, and the assistant’s ask/guess policy is chosen (explicitly, or as an emergent property of training and product incentives) with some implicit model of how users will respond.

This paper provides a mechanism-design theory of endogenous clarification in AI assistance. Its central premise is that strategic under-specification is not caused by AI getting better at guessing; it is caused by users responding to the assistant’s endogenous clarification policy when the assistant’s objective differs from theirs. Section 2 begins with a decision-theoretic benchmark where the assistant’s policy is exogenous, proving that improved guessing accuracy alone weakly improves user welfare despite shortening prompts. Section 3 identifies the true moral-hazard channel: unobserved divergence between the user’s belief and the assistant’s true asking policy. Sections 4–7 close the loop by modeling the assistant as a Stackelberg leader anticipating user best responses under pooling and screening instruments. Section 8 translates the equilibrium characterizations into seven falsifiable behavioral predictions (P1–P7). Section 10 evaluates exact nonlinear robustness and discusses institutional extensions.

1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

Contract theory and screening without transfers. Our screening framework builds on classical adverse-selection models (Myerson, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate a offered in response to specification effort m). This connects to non-monetary screening and costly verification (Townsend, 1979), where information revelation is governed by operational friction rather than cash transfers.

Delegation, authority, and information design. Our analysis relates closely to the theory of delegation under asymmetric information (Holmström, 1984), where a principal determines how much discretion to grant an agent, and to models of formal versus real authority (Aghion and Tirole, 1997), where an agent’s upstream effort is shaped by who retains authority over downstream implementation. In our setting, leaving attributes unspecified implicitly delegates their resolution to the assistant. Furthermore, because the assistant’s asking policy determines which state dimensions are actively resolved, our framework shares conceptual foundations with Bayesian persuasion and strategic information design (Kamenica and Gentzkow, 2011), while focusing on the moral-hazard trade-off in prompt articulation.

Costly communication and strategic disclosure. The user’s specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user’s rational economizing on specification costs when anticipating automated assistance.

Assistance games and human–AI interaction. Our bi-level interaction connects to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al.,

2016; von Stackelberg, 1934), where an autonomous assistant acts to maximize human utility under incomplete information. However, while assistance games formalize the assistant’s uncertainty over the human’s underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value V and error loss L). Instead, the informational asymmetry resides entirely in the user’s private *specification cost* κ . The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. Our model provides a contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

Clarification and interactive NLP. A parallel empirical literature in natural language processing and information retrieval investigates algorithms for generating and ranking clarifying questions in conversational search and dialog (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020). These works optimize static clarification decisions or train heuristic dialog policies via reinforcement learning against fixed datasets. However, they uniformly treat user inputs as exogenous test instances rather than the output of an optimizing agent who strategically anticipates the clarification policy. Our framework provides the contract-theoretic foundation that this engineering literature lacks, formally endogenizing prompt specification effort as an equilibrium best response to the assistant’s asking policy.

2 Environment and the Decision-Theoretic Benchmark

2.1 Setup

A task has k binary, independent, intent-relevant attributes $\theta = (\theta_1, \dots, \theta_k) \in \{0, 1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in \{0, \dots, k\}$ and truthfully reveals m attributes.¹ Specifying m attributes costs $c(m)$, with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$.

For each of the $k - m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0, 1)$. We model the assistant’s policy at this stage as a uniform ask rate $a \in [0, 1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a, g) = a + (1 - a)g$. Task success requires *all* k attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user’s payoff is

$$U(m; a, g) = V q(a, g)^{k-m} - c(m) - a c_Q (k - m),$$

with $V > 0$ the value of success and $c_Q > 0$ the user’s cognitive cost of answering one clarifying question. We treat m as continuous on $[0, k]$ for tractability.

2.2 Proposition 1: accuracy alone does not create moral hazard

Proposition 1. *Fix a . Let $U^*(g) = \max_{m \in [0, k]} U(m; a, g)$. Then U^* is weakly increasing in g .*

Proof. By the envelope theorem, at the optimum $m^*(g)$, $dU^*/dg = \partial U / \partial g|_{m=m^*(g)} = V(k - m^*)q^{k-m^*-1}(1 - a) \geq 0$. The envelope theorem lets us ignore dm^*/dg : because m^* is chosen optimally, its first-order response to g has no first-order effect on U^* . $\square$

¹We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

Assumption 1 (Regularity). *In the relevant range of m , $(k - m) \ln q(a, g) \geq -1$.*

Remark 1 (Interpreting Assumption 1 and Model Concavity). *The regularity condition $(k - m) \ln q(a, g) \geq -1$ is a sufficient condition for the single-crossing property used in Corollary 1 via Topkis’s monotone comparative statics theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994). The direction is pinned down by the cross-partial of utility: for fixed a ,*

$$\frac{\partial^2 U}{\partial m \partial g} = -V(1 - a)q^{k-m-1}[(k - m) \ln q + 1].$$

Thus U exhibits decreasing differences in (m, g) , which Topkis’s theorem requires to guarantee that $m^(g)$ is weakly decreasing, if and only if $(k - m) \ln q + 1 \geq 0$, which is precisely Assumption 1. Since $\ln q \approx -(1 - q)$ for q near 1, this condition requires $(k - m)(1 - q) \lesssim 1$: the expected number of residual unresolved attributes under guessing is at most about one. If this condition fails ($(k - m) \ln q < -1$), U exhibits increasing differences, and higher guessing accuracy can locally induce higher specification effort before falling as $q \rightarrow 1$.*

We also note that in the exact conjunctive model, the gross benefit Vq^{k-m} is strictly convex in m (since $\partial^2[Vq^{k-m}]/\partial m^2 = Vq^{k-m}(\ln q)^2 > 0$). Global concavity of U in m therefore requires the specification cost curvature $c''(m) = \kappa$ to strictly exceed $Vq^{k-m}(\ln q)^2$ across $[0, k]$. Under the linear-risk specification introduced in Section 4, by contrast, gross benefit is linear in m , guaranteeing that U is globally strictly concave in m for all $\kappa > 0$.

Corollary 1. *Under Assumption 1, $m^*(g)$ is weakly decreasing in g (decreasing differences in (m, g) ; Topkis’s theorem).*

Proof. Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition $\partial^2 U / \partial m \partial g \leq 0$ on the relevant range, which is decreasing differences in (m, g) ; Topkis’s theorem then gives $m^*(g)$ weakly decreasing. $\square$

Thus, holding the assistant’s policy fixed, users specify less as the assistant becomes more adept at guessing, yet they are never worse off for doing so. Proposition 1 serves primarily as a baseline: it applies the envelope theorem to a single agent optimizing against a technology parameter that enters the payoff with a non-negative derivative, a property that holds generally in models of this class. Its purpose is not to present a surprising result, but to establish the rigorous benchmark against which claimed moral hazard must be evaluated, ensuring that Proposition 2 isolates the true source of welfare loss. This frames the central question of this paper: what happens once the policy a is no longer exogenous, but chosen by a strategic assistant that anticipates the user’s response?

3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant’s ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

3.1 Cost externality

If a third party (such as a code reviewer or downstream production system) bears part of the cost of an erroneous guess, the user’s private parameters (κ, V) understate the social costs, causing the user to under-specify relative to the social optimum at any fixed g , an inefficiency that worsens as g rises by Corollary 1. While this constitutes a standard Pigouvian externality, our primary focus lies in strategic and informational friction.

3.2 Miscalibrated belief updating under silent failure

If incorrect guesses are not always detected, and the detection probability $d(m)$ is increasing in m , users infer g from experience rather than observing it directly. Under-specifying users are consequently less likely to catch their own errors, generating an upward-biased estimate $\hat{g}$ that reinforces under-specification. This mechanism represents a dynamic learning process rather than a static equilibrium outcome; we formalize its testable implications in Section 8 (prediction P5).

3.3 Belief about the assistant’s policy and equilibrium under-specification

When the clarification policy a is endogenous, the user’s specification decision depends on their belief regarding the assistant’s asking rate. Let users be *sophisticated* (correctly inferring the assistant’s true operational ask rate a_A) or *naive* (believing the assistant follows a higher clarification rate $a^\dagger > a_A$, such as the unconstrained welfare-optimal policy).

Definition 1 (Strategic Under-Specification). *A user exhibits strategic under-specification if their chosen prompt specification effort m is strictly lower than the full-information, correctly calibrated benchmark:*

- (i) *Relative to sophisticated belief calibration: $m_{naive}^* < m_{soph}^*$; or*
- (ii) *Relative to unassisted specification: $m^*(a) < m^*(0)$ under automated assistance.*

In human–AI interaction, varying the clarification rate a engages two opposing economic channels:

1. *The accuracy channel:* A higher ask rate increases effective resolution accuracy $q(a, g) = a + (1 - a)g$. Under decreasing differences (Assumption 1), greater accuracy reduces the marginal value of upstream prompt effort, pushing specification downward.
2. *The question-cost channel:* Each clarifying question imposes an interactive answering cost c_Q on the user. Because the total answering cost $ac_Q(k - m)$ is decreasing in specification m , a higher ask rate increases the marginal penalty of leaving attributes unspecified, pushing specification upward (pre-emption).

Under the linear-risk specification analyzed in Sections 4–6, where the net benefit of clarification $s \equiv \Lambda - c_Q > 0$ is positive, the cross-partial derivative of user utility is:

$$\frac{\partial^2 U}{\partial m \partial a} = -(\Lambda - c_Q) < 0.$$

The accuracy channel strictly dominates the question-cost channel, ensuring that U exhibits strictly decreasing differences in (m, a) .

Proposition 2 (Equilibrium Under-Specification under Naive Beliefs). *Under the linear-risk specification with $\Lambda > c_Q$, naive users choose strictly lower specification effort than sophisticated users,*

$$m_{naive}^* < m_{soph}^*,$$

and achieve strictly lower true welfare: $U(m_{naive}^; a_A) < U(m_{soph}^*; a_A)$. Holding raw guessing accuracy g fixed, naive users strategically under-specify and suffer an equilibrium welfare loss.*

Proof. Under linear risk, the follower’s best response to an anticipated ask rate a is $m^*(\kappa; a) = \frac{\Lambda(1-a)+ac_Q}{\kappa}$. Since $\Lambda > c_Q$, differentiating with respect to a gives $\partial m^*/\partial a = (c_Q - \Lambda)/\kappa < 0$. A naive user anticipating $a^\dagger > a_A$ therefore chooses:

$$m_{\text{naive}}^* = m^*(\kappa; a^\dagger) < m^*(\kappa; a_A) = m_{\text{soph}}^*.$$

Because true utility $U(m; a_A)$ is strictly concave in m with unique global maximizer at m_{soph}^* , any choice $m \neq m_{\text{soph}}^*$ incurs a strict utility loss. Evaluating the quadratic shortfall directly:

$$U(m_{\text{soph}}^*; a_A) - U(m_{\text{naive}}^*; a_A) = \frac{\kappa}{2} (m_{\text{soph}}^* - m_{\text{naive}}^*)^2 = \frac{(\Lambda - c_Q)^2}{2\kappa} (a^\dagger - a_A)^2 > 0,$$

which proves that naive users are strictly worse off. $\square$

Remark 2 (Exact Model Counterexample to Monotone Under-Specification). *In the exact conjunctive model $U(m, a) = Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the cross-partial derivative is:*

$$\frac{\partial^2 U}{\partial m \partial a} = -V(1 - g) q^{k-m-1} [(k - m) \ln q + 1] + c_Q.$$

Even when Assumption 1 holds, the question-answering cost term $+c_Q$ can dominate if questions are costly relative to baseline task value V . For example, with $k = 5, V = 10, \kappa = 0.5, g = 0.8, c_Q = 2, a_A = 0.1$, and $a^\dagger = 0.6$, Assumption 1 holds across the domain, yet sophisticated users choose $m_{\text{soph}}^ = 3.15$ while naive users choose $m_{\text{naive}}^* = 3.93$. Here, the naive user anticipates substantial clarification costs and over-specifies upstream to avoid conversational interruptions. Proposition 2 thus isolates the regime where clarification is socially cost-effective ($\Lambda > c_Q$), which is the relevant environment for helpful automated assistants.*

This identifies the first true source of moral hazard: not technological accuracy, but an unobserved gap between the user’s belief regarding the assistant’s policy and the assistant’s actual policy. Section 5 solves for the optimal policy a_A when the assistant acts as a strategic, potentially biased Stackelberg leader, allowing Proposition 2 to be evaluated with an endogenously derived policy rather than an exogenously assumed one.

4 The Assistant’s Policy as a Stackelberg Game

4.1 General formulation

The assistant (leader) commits ex ante to a policy π before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m; \kappa) = \kappa m^2/2$, and best-responds $m^*(\kappa; \pi) = \arg \max_m U(m; \pi, \kappa)$. The leader’s own objective Π need not equal U : we write

$$\Pi_\kappa(m, a) = \mu_A U(m, a; \kappa) - (\lambda_A - c_Q) a (k - m),$$

where $\mu_A \in (0, 1]$ discounts the leader’s weight on true downstream user welfare, c_Q is the user’s true cognitive cost of answering one clarifying question, and λ_A is the leader’s own perceived or institutional cost per clarifying question. In commercial AI systems, misalignment along these margins arises from concrete engineering and economic forces:

1. *Asking friction bias* ($\lambda_A > c_Q$): Reinforcement learning from human feedback (RLHF) and preference optimization algorithms (e.g., DPO) commonly evaluate conversation turns in isolation.

Reward models systematically penalize “evasive” clarifying questions in favor of immediate code generation because users in rating pools favor direct answers. Furthermore, multi-turn interactions increase token costs, context-window expansion, and inference latency, creating provider-side pressure to suppress clarification turns.

2. *Downstream welfare discounting* ($\mu_A < 1$): Turn-level user satisfaction signals (such as prompt acceptance or immediate positive feedback) capture immediate session fluency. In contrast, the true economic loss L from broken conventions, subtle edge-case errors, or security defects is typically discovered hours or days later in downstream execution, test suites, or code review. Consequently, training objectives tuned to immediate session retention tend to discount downstream correctness.

The leader is *unbiased* iff $\mu_A = 1$ and $\lambda_A = c_Q$, in which case $\Pi_\kappa \equiv U(\cdot; \kappa)$ exactly. The leader solves $\pi^* = \arg \max_\pi \mathbb{E}_{\kappa \sim F} [\Pi_\kappa(m^*(\kappa; \pi), \pi)]$, a bilevel program.

Remark 3 (Existence). *For Model I below ($\pi = a \in [0, 1]$, a compact scalar policy), $m^*(\kappa; a)$ is continuous in a (strict concavity of U in m plus the implicit function theorem), so $\mathbb{E}_\kappa[\Pi_\kappa]$ is continuous in a and a maximizer exists on $[0, 1]$ by Weierstrass’s theorem. For Model II below (a finite menu over two types), the feasible set is a compact subset of $\mathbb{R}^4$ cut out by linear IC/IR constraints and a continuous objective, so a maximizer again exists by Weierstrass’s theorem; no ironing or ex-ante randomization is needed with only two types.*

For tractability in what follows we replace the exact conjunctive success probability q^{k-m} with its first-order (small-risk) approximation around $q \rightarrow 1$: expected loss $\approx L(1 - q)(k - m)$, where L denotes the economic loss from a single unresolved-and-incorrect attribute. With $\Lambda \equiv L(1 - g)$, the user’s payoff becomes

$$U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - a c_Q (k - m).$$

Throughout Sections 4.2–6, we maintain the physical feasibility constraint $m \in [0, k]$, so that followers cannot specify more attributes than exist in the task. When unconstrained first-order conditions yield $m^*(\kappa; a) = \frac{\Lambda(1-a)+ac_Q}{\kappa} > k$ (which occurs for low-cost types with $\kappa < \Lambda/k$), best-response effort saturates at the physical boundary $m^* = k$.

4.2 The first-best benchmark and physical feasibility

Before analyzing either strategic regime, we characterize the fully-informed, fully-aligned social optimum:

$$(m^{FB}(\kappa), a^{FB}(\kappa)) = \arg \max_{m \in [0, k], a \in [0, 1]} U(m, a; \kappa).$$

Proposition 3 (First-Best Policy on the Feasible Domain). *On the physically feasible domain $m \in [0, k]$, under the linear-risk specification with $\Lambda > c_Q$:*

- (i) *Asking weakly dominates guessing for every cost type $\kappa > 0$: $a^{FB}(\kappa) = 1$, with optimal specification $m^{FB}(\kappa) = \min\{k, c_Q/\kappa\}$.*
- (ii) *For any type with $\kappa > c_Q/k$, asking strictly dominates guessing: $U(m^*(1), 1) > U(m^*(0), 0)$. For boundary types with $\kappa \leq c_Q/k$, specification effort saturates at $m^{FB} = k$, leaving zero residual attributes ($k - m = 0$) and rendering the asking policy payoff-irrelevant.*

Proof. For any fixed specification effort $m \in [0, k]$, the partial derivative of utility with respect to a is:

$$\frac{\partial U}{\partial a} = (\Lambda - c_Q)(k - m).$$

Because $\Lambda > c_Q$, this derivative is strictly positive for all $m < k$ and zero at $m = k$. Now consider the joint optimization over $m \in [0, k]$ and $a \in \{0, 1\}$. Under $a = 1$, the user chooses $m^*(1) = \min\{k, c_Q/\kappa\}$. Under $a = 0$, the user chooses $m^*(0) = \min\{k, \Lambda/\kappa\}$. We evaluate the three type intervals:

- If $\kappa \geq \Lambda/k$, both best responses remain interior: $m^*(1) = c_Q/\kappa \leq m^*(0) = \Lambda/\kappa \leq k$. Evaluating the welfare difference:

$$U(m^*(1), 1; \kappa) - U(m^*(0), 0; \kappa) = (\Lambda - c_Q) \left(k - \frac{\Lambda + c_Q}{2\kappa} \right).$$

Since $\kappa \geq \Lambda/k > \frac{\Lambda + c_Q}{2k}$, the bracketed term satisfies $k - \frac{\Lambda + c_Q}{2\kappa} > k - k = 0$. Hence $U(m^*(1), 1; \kappa) > U(m^*(0), 0; \kappa)$ strictly.

- If $c_Q/k < \kappa < \Lambda/k$, unconstrained effort under $a = 0$ exceeds k , so specification under guessing saturates at the boundary $m^*(0) = k$, yielding $U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$. Under asking, effort remains interior at $m^*(1) = c_Q/\kappa < k$, yielding $U(m^*(1), 1; \kappa) = V - c_Qk + \frac{c_Q^2}{2\kappa}$. Subtracting:

$$U(m^*(1), 1; \kappa) - U(k, 0; \kappa) = \frac{\kappa k^2}{2} - c_Qk + \frac{c_Q^2}{2\kappa} = \frac{(\kappa k - c_Q)^2}{2\kappa} > 0.$$

- If $\kappa \leq c_Q/k$, specification effort saturates at $m^* = k$ under both policies, so $U(k, 1; \kappa) = U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$.

Therefore, $a^{FB}(\kappa) = 1$ is globally optimal across all types $\kappa > 0$, with indifference only on the boundary set $\kappa \leq c_Q/k$ where $m^{FB} = k$. $\square$

Remark 4 (The Unclipped Threshold κ^* and Genuine Asking–Guessing Trade-offs). *In unconstrained algebra without the feasibility constraint $m \leq k$, setting the difference $(\Lambda - c_Q)(k - \frac{\Lambda + c_Q}{2\kappa})$ to zero yields the formal threshold $\kappa^* \equiv \frac{\Lambda + c_Q}{2k}$. However, since $\Lambda > c_Q$, we have $\kappa^* < \Lambda/k$ strictly. The region $\kappa < \kappa^*$ where guessing formally appeared to beat asking in unclipped algebra lies entirely in the unphysical domain where unconstrained effort Λ/κ exceeds k (the user would specify more attributes than exist in the task). Once $m \leq k$ is enforced, asking weakly dominates guessing everywhere under linear risk because asking resolves uncertainty at lower unit cost ($c_Q < \Lambda$) while allowing the user to economize on specification effort.*

Genuine interior thresholds where $a^{FB} \in \{0, 1\}$ strictly partitions the type space arise from two natural richer specifications:

1. *Exact conjunctive success: Under $Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the marginal gain from clarification is $(k - m)[V(1 - g)q^{k-m-1} - c_Q]$. When baseline accuracy g is high or $k - m$ is small, this marginal gain turns negative, making guessing strictly optimal for low-cost types who specify most attributes upstream.*
2. *Heterogeneous attribute stakes: If attributes vary in importance with stakes Λ_i , efficiency prescribes asking only on high-stakes attributes ($\Lambda_i > c_Q$) and guessing on low-stakes attributes ($\Lambda_i < c_Q$).*

5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0, 1]$ and cannot condition on user-specific attributes or observed effort. This provides a natural benchmark for policies embedded uniformly at training time or applied without user-level profiling.

5.1 User best response

The FOC $\partial U / \partial m = \Lambda(1 - a) - \kappa m + a c_Q = 0$ gives, with $\beta(\kappa) = \Lambda / \kappa$ and $\gamma(\kappa) = (\Lambda - c_Q) / \kappa$,

$$m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa), \quad \frac{\partial m^*}{\partial a} = -\gamma(\kappa) = \frac{c_Q - \Lambda}{\kappa}.$$

We assume interior specification levels $m^* \in (0, k)$; for low-cost types where unconstrained $\beta(\kappa) \geq k$, specification effort saturates at $m^* = k$. The closed-form population moments $\bar{R}_0$ and $\bar{\gamma}$ below are exact when best responses remain interior, and serve as tractable linear approximations when boundary clipping binds; as verified in Section 10 and our computational robustness suite, the qualitative regime structure and corner-solution properties are strictly preserved under boundary clipping and exact payoffs alike.

Remark 5 (Non-monotone effect of ask rate on specification). *Unlike Corollary 1, the sign of $\partial m^* / \partial a$ here does not depend on type: it is positive for every type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that condition holds, a higher population ask rate induces users to specify more, not less; they pre-empt costly clarifying questions by articulating the answer directly. This introduces a new economic force relative to Section 2, where a operated solely through effective accuracy, and generates a distinct, directly testable prediction (prediction P1' below).*

5.2 The leader's problem

Let $R(a, \kappa) = k - m^*(\kappa; a)$ denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate $a \in [0, 1]$ applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.2 at the follower's optimal best response $m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa)$:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A U(m^*(\kappa; a), a; \kappa) - (\lambda_A - c_Q) a (k - m^*(\kappa; a)).$$

Substituting $U(m^*(\kappa; a), a; \kappa) = V - \Lambda(1 - a)(k - m^*) - \frac{\kappa}{2}(m^*)^2 - a c_Q (k - m^*)$, using $k - m^*(\kappa; a) = R_0(\kappa) + a \gamma(\kappa)$, and defining the effective resolution stake $s \equiv \Lambda - c_Q$ and the assistant's excess asking friction $b \equiv \lambda_A - c_Q$, expanding in powers of a yields:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A \left(V - \Lambda k + \frac{\Lambda^2}{2\kappa} \right) + (\mu_A s - b) R_0(\kappa) a + \frac{(\mu_A s - 2b) \gamma(\kappa)}{2} a^2.$$

Averaging over $\kappa \sim F$ requires no moments beyond the population sufficient statistics already defined, $\bar{R}_0 = k - \mathbb{E}[\beta]$ and $\bar{\gamma} = \mathbb{E}[\gamma] = s \mathbb{E}[1/\kappa]$:

$$\Pi(a) = \text{const} + \bar{L} a + \bar{Q} a^2,$$

where $\text{const} = \mu_A (V - \Lambda k) + \frac{\mu_A \Lambda^2}{2} \mathbb{E}[1/\kappa]$, and

$$\bar{L} \equiv (\mu_A s - b) \bar{R}_0, \quad \bar{Q} \equiv \frac{(\mu_A s - 2b) \bar{\gamma}}{2}.$$

Proposition 4 (Stackelberg-optimal pooling rate). *Provided $\bar{Q} < 0$ (the second-order condition for strict concavity, equivalent to $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$), the leader's unconstrained stationary point is*

$$a^{SE} = -\frac{\bar{L}}{2\bar{Q}} = -\frac{(\mu_A s - b)\bar{R}_0}{(\mu_A s - 2b)\bar{\gamma}}.$$

When this stationary point lies in $[0, 1]$, it is the unique global maximum of $\Pi(a)$ on $[0, 1]$.

Corollary 2 (Friction bias suppresses the pooling ask rate on the interior branch). *On the interior branch $\bar{Q} < 0$, differentiating the closed form in Proposition 4 directly with respect to λ_A (holding primitives $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$ fixed) gives:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A s}{(\mu_A s - 2b)^2} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A(\Lambda - c_Q)}{[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)]^2}.$$

On the physically valid domain where best responses satisfy $m^(\kappa) \leq k$, average residual unspecified attributes satisfy $\bar{R}_0 = k - \mathbb{E}[m^*(0)] \geq 0$. Because $\bar{\gamma} > 0$, $\mu_A > 0$, and $s > 0$, it follows that:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} \leq 0.$$

Higher perceived clarification friction λ_A strictly suppresses the assistant's equilibrium asking rate on the interior margin.

Corollary 3 (Non-identification of the two bias channels in the interior regime). *On the interior branch ($\bar{Q} < 0$), defining the normalized excess friction per unit of altruism $\tilde{\rho} \equiv \frac{\lambda_A - c_Q}{\mu_A} = \frac{b}{\mu_A}$, the optimal pooling rate simplifies to*

$$a^{SE} = -\frac{\bar{R}_0}{\bar{\gamma}} \left[\frac{s - \tilde{\rho}}{s - 2\tilde{\rho}} \right],$$

so a^{SE} depends on the two bias parameters (μ_A, λ_A) strictly through the single scalar ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$ (holding primitives s and $\bar{R}_0/\bar{\gamma}$ fixed). Consequently,

$$\frac{\partial a^{SE}/\partial \mu_A}{\partial a^{SE}/\partial \lambda_A} = -\frac{\lambda_A - c_Q}{\mu_A} = -\tilde{\rho},$$

and any two parameter pairs on the same ray $(\lambda_A - c_Q)/\mu_A = \text{const}$ produce the identical pooling ask rate. A leader that discounts downstream welfare ($\mu_A < 1$, $\lambda_A = c_Q \implies \tilde{\rho} = 0$) cannot be distinguished, from a^{SE} alone, from any leader along the corresponding ray of excess friction.

Remark 6 (Scope and structure of Corollary 3). *This ray-invariance arises because the welfare discount μ_A scales the user's entire objective U in the leader formulation $\Pi = \mu_A U - (\lambda_A - c_Q)a(k - m)$. If the provider instead discounted only downstream task loss L while internalizing user interaction costs c_Q at full weight, μ_A and λ_A would enter through independent marginal channels. Furthermore, by Corollary 5 below, the unbiased pooling optimum is always a corner ($\bar{Q} \geq 0$), so the interior stationary point describes behavior when friction bias is sufficiently large ($2b > \mu_A s$).*

Corollary 4 (Ray-invariance extends to the corner regime). *On the corner branch ($\bar{Q} \geq 0$, or whenever the unconstrained stationary point lies outside $[0, 1]$), the optimal policy is chosen by the sign of the corner difference $\Pi(1) - \Pi(0) = \bar{L} + \bar{Q}$:*

$$\Pi(1) - \Pi(0) = \mu_A \left[s \left(\bar{R}_0 + \frac{\bar{\gamma}}{2} \right) - \tilde{\rho}(\bar{R}_0 + \bar{\gamma}) \right].$$

Since $\mu_A > 0$, μ_A factors out completely, and the sign depends solely on the scalar ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$. The critical threshold ratio is

$$\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}},$$

with $a^{SE} = 1$ for $\tilde{\rho} < \tilde{\rho}^*$ and $a^{SE} = 0$ for $\tilde{\rho} > \tilde{\rho}^*$. Ray-invariance in the excess friction ratio $\tilde{\rho}$ therefore holds unconditionally across the entire parameter space, interior and corner regimes alike.

Corollary 5 (Pooling is a corner solution in the unbiased case). *In the unbiased case ($\mu_A = 1, \lambda_A = c_Q \implies b = 0$), the quadratic coefficient is:*

$$\bar{Q} = \frac{s^2 \bar{\gamma}}{2} = \frac{(\Lambda - c_Q)^2 \mathbb{E}[1/\kappa]}{2} \geq 0 \quad \text{always.}$$

The second-order condition for an interior maximum therefore fails: $\Pi(a)$ is weakly convex on $[0, 1]$, its stationary point is a local minimum, and the true optimum is always a corner, $a^{SE} \in \{0, 1\}$, not an interior compromise. Comparing the two corners directly in unclipped algebra:

$$\Pi(1) - \Pi(0) = \bar{L} + \bar{Q} = s \left(\bar{R}_0 + \frac{\bar{\gamma}}{2} \right) = (\Lambda - c_Q) \left[k - \frac{\Lambda + c_Q}{2} \mathbb{E} \left[\frac{1}{\kappa} \right] \right],$$

which yields the formal unclipped threshold $k^* \equiv \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$. When follower best responses are physically constrained to the feasible domain $m \leq k$, Proposition 3 implies that asking ($a = 1$) weakly dominates guessing for every individual type in the support, so unbiased pooling under physical feasibility selects $a^{SE} = 1$ unconditionally. Genuine interior trade-offs between asking and guessing at the population level emerge under the exact conjunctive success model, as analyzed in Section 10.

This property proves essential for the comparative analysis in Section 7: Model I exhibits welfare loss relative to the first best primarily through objective bias, which shifts a^{SE} (Corollary 2) and suppresses the operational clarification rate.

6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types $\kappa_L < \kappa_H$ with population shares f_L, f_H (the continuum-type extension using the standard Mirrlees technique is discussed in Section 10). Let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \quad \text{IR}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$.

6.1 Single-crossing

The marginal rate of substitution between m and a along an indifference curve is $\text{MRS}(m, a; \kappa) = -(\partial U / \partial m) / (\partial U / \partial a)$. From user utility $U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2} m^2 - ac_Q(k - m)$,

we have $\partial U/\partial m = \Lambda(1 - a) + ac_Q - \kappa m$ and $\partial U/\partial a = s(k - m)$, where $s \equiv \Lambda - c_Q > 0$ is the net social gain from asking. Hence,

$$\text{MRS}(m, a; \kappa) = \frac{\kappa m - [\Lambda - as]}{s(k - m)}, \quad \frac{\partial \text{MRS}}{\partial \kappa} = \frac{m}{s(k - m)} > 0,$$

which holds strictly for all interior specification levels $0 < m < k$ under the maintained condition $s > 0$. Higher- κ types therefore have strictly steeper indifference curves in (m, a) -space: they require more compensating ask-service per unit of specification effort given up. This satisfies the standard Spence–Mirrlees single-crossing condition (Spence, 1973) and underwrites the screening analysis below.

6.2 No distortion without bias

Proposition 5 (Efficiency survives private information when the leader is unbiased). *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot; \kappa)$ for every κ), then implementing each type’s own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

Proof. $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot; \kappa)$ by definition, so for any $\kappa' \neq \kappa$, $U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \geq U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$ trivially: no bundle can give type κ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot; \kappa)$, the leader’s payoff from serving each true type is maximized at that type’s own first best, so this menu is also leader-optimal. $\square$

This result highlights a fundamental distinction between human–AI coordination and classical adverse selection (such as Baron–Myerson or Laffont–Tirole). In regulatory and auction models, the principal pays monetary transfers to extract surplus and thus treats informational rents as a pure loss. Here, by contrast, the leader’s objective can coincide directly with user welfare; when it does, incentive compatibility is costless, and private information incurs no welfare penalty. Distortion in this setting is therefore a pure signature of objective misalignment, not of asymmetric information per se.

6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot; \kappa)$ (bias present), the leader’s preferred bundle per type,

$$(m_\kappa^B, a_\kappa^B) = \arg \max_{m, a} \Pi_\kappa(m, a)$$

(computable in closed form by the same method as Proposition 3, with Λ, c_Q replaced by their μ_A, λ_A -weighted counterparts), no longer automatically satisfies IC, because Π_κ and the user’s true $U(\cdot; \kappa)$ now rank bundles differently.

Proposition 6 (Downward distortion of the high-cost type’s service). *Assume $\Lambda > c_Q$, $\kappa_L < \kappa^{*B} < \kappa_H$, under-asking bias ($\lambda_A > \mu_A c_Q$, so $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A > c_Q$), and task complexity $k > k_{crit}$ (defined in (1)). Then:*

1. **Incentive violation of unconstrained menu:** *The unconstrained biased menu $(m_L^B, a_L^B) = (\min\{k, \Lambda/\kappa_L\}, 0)$ and $(m_H^B, a_H^B) = (\min\{k, c_Q^{eff}/\kappa_H\}, 1)$ strictly violates IC_L whenever the type heterogeneity spread $\kappa_H - \kappa_L$ is sufficiently large.*

2. **Downward distortion:** Under the optimal incentive-compatible menu, the low-cost type receives no distortion ($a_L^{SB} = a_L^B = 0$), while the high-cost type's ask rate is strictly distorted downward:

$$a_H^{SB} < a_H^B \leq a_H^{FB} = 1.$$

3. **Active constraint regime:** In the moderate-bias regime, IC_L binds alone while IR_H , IC_H , and IR_L remain strictly slack. Under sufficiently extreme bias, IC_H can become co-active with IC_L , and the optimal menu may collapse to a pooling or boundary-saturated allocation.

Proof. We establish the result in five analytical steps:

Step 1: Unconstrained biased optima. The leader objective per type is $\Pi_\kappa(m, a) = \mu_A[V - \Lambda(1-a)(k-m) - \frac{\kappa}{2}m^2 - ac_Q^{eff}(k-m)]$, where $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A > c_Q$ in the presence of under-asking bias. Maximizing unconstrained with respect to m yields $m^B(\kappa; a) = [\Lambda(1-a) + ac_Q^{eff}]/\kappa$. Because Π_κ is linear in a for fixed m with slope $\mu_A(\Lambda - c_Q^{eff})(k-m)$, the biased first-best ask rate is bang-bang around the threshold $\kappa^{*B} = (\Lambda + c_Q^{eff})/(2k)$. For $\kappa_L < \kappa^{*B} < \kappa_H$, the unconstrained menu is $(m_L^B, a_L^B) = (\min\{k, \Lambda/\kappa_L\}, 0)$ and $(m_H^B, a_H^B) = (\min\{k, c_Q^{eff}/\kappa_H\}, 1)$.

Step 2: Upward temptation to mimic (IC_L violation). Evaluating type κ_L 's payoff from accepting the high type's bundle $(m_H^B, 1)$ versus its own bundle $(m_L^B, 0)$:

$$U(m_H^B, 1; \kappa_L) - U(m_L^B, 0; \kappa_L) = \frac{\kappa_H - \kappa_L}{2}(m_H^B)^2 > 0.$$

Because type κ_L can provide the high type's specification level $m_H^B = c_Q^{eff}/\kappa_H$ at strictly lower cognitive cost ($\kappa_L < \kappa_H$), its utility from mimicking type H exceeds type H 's own utility by the informational rent $\frac{\kappa_H - \kappa_L}{2}(m_H^B)^2$. When $\kappa_H - \kappa_L$ is sufficiently wide, $U(m_H^B, 1; \kappa_L) > U(m_L^B, 0; \kappa_L)$, violating IC_L .

Step 3: Slackness of participation (IR_H). In classical adverse selection with monetary transfers (Baron–Myerson, Laffont–Tirole), the principal's payoff is linearly reduced by agent rent, driving the inefficient agent's IR to bind at reservation $\underline{U}$. Here, the principal's payoff Π_{κ_H} directly incorporates $\mu_A U(\cdot; \kappa_H)$. Depressing U_H directly penalizes the leader's own objective. Thus, the leader adjusts the bundle (m_H, a_H) only as far as necessary to restore IC_L , leaving IR_H strictly slack: $U(m_H^{SB}, a_H^{SB}; \kappa_H) > \underline{U}$.

Step 4: Global optimization along the IC_L boundary. When IC_L binds, any feasible allocation must satisfy $U(m_H, a_H; \kappa_L) \leq U(m_L, a_L; \kappa_L)$. First, consider the low-cost bundle (m_L, a_L) . Because $(m_L^B, a_L^B) = (\min\{k, \Lambda/\kappa_L\}, 0)$ is the unconstrained maximizer of both $U(\cdot; \kappa_L)$ and Π_{κ_L} on the feasible domain, standard no distortion at the top applies: $(m_L^{SB}, a_L^{SB}) = (m_L^B, 0)$, fixing type L 's utility at its reservation level $U_L^0 \equiv U(m_L^B, 0; \kappa_L)$. When $\Lambda/\kappa_L \geq k$ (as in our numerical calibration where $\Lambda/\kappa_L = 16.67 > k = 10$), type L 's specification effort saturates at $m_L^B = k$, leaving zero unspecified attributes ($k - m_L^B = 0$); asking service a_L is therefore payoff-irrelevant to type L , yielding reservation utility $U_L^0 = V - \frac{\kappa_L}{2}k^2$. Setting $a_L^{SB} = 0$ is costless for type L and minimizes the attractiveness of type L 's bundle to type H . To implement this direct mechanism when the

follower chooses specification effort m , the assistant specifies an off-menu deterrence schedule:

$$a(m) = \begin{cases} a_H^{SB}, & \text{if } m = m_H^{SB}, \\ a_L^{SB} = 0, & \text{if } m = m_L^{SB}, \\ 0, & \text{if } m \notin \{m_L^{SB}, m_H^{SB}\}. \end{cases}$$

Any deviation to an off-menu choice $m \notin \{m_L^{SB}, m_H^{SB}\}$ receives zero clarification ($a = 0$) and is strictly dominated for both types.

The entire burden of satisfying IC_L must therefore fall on the high-cost bundle (m_H, a_H) . Along the binding boundary $U(m_H, a_H; \kappa_L) = U_L^0$, utility is strictly linear in a_H :

$$V - \Lambda(k - m_H) + a_H s(k - m_H) - \frac{\kappa_L}{2} m_H^2 = U_L^0,$$

where $s \equiv \Lambda - c_Q > 0$. Because $s(k - m_H) > 0$ for all $m_H < k$, this uniquely determines the required asking rate a_H as a function of specification effort m_H :

$$a_H(m_H)(k - m_H) = \frac{1}{s} \left[U_L^0 - V + \Lambda(k - m_H) + \frac{\kappa_L}{2} m_H^2 \right],$$

or equivalently,

$$a_H(m_H) = 1 - \frac{U(m_H, 1; \kappa_L) - U_L^0}{s(k - m_H)}.$$

Next, evaluate the leader's payoff $\Pi_{\kappa_H}(m_H, a_H) = \mu_A U(m_H, a_H; \kappa_H) - b a_H(k - m_H)$ along this boundary, where $b \equiv \lambda_A - c_Q > 0$. Using the rent identity $U(m_H, a_H; \kappa_H) = U_L^0 - \frac{\kappa_H - \kappa_L}{2} m_H^2$ and substituting $a_H(m_H)(k - m_H)$, the leader's reduced objective becomes an exact function of m_H alone:

$$\Pi_{\kappa_H}(m_H) = \mu_A \left[U_L^0 - \frac{\kappa_H - \kappa_L}{2} m_H^2 \right] - \frac{b}{s} \left[U_L^0 - V + \Lambda(k - m_H) + \frac{\kappa_L}{2} m_H^2 \right].$$

Expanding terms:

$$\Pi_{\kappa_H}(m_H) = \left(\mu_A - \frac{b}{s} \right) U_L^0 + \frac{b}{s} (V - \Lambda k) + \frac{b \Lambda}{s} m_H - \frac{1}{2} \left[\mu_A (\kappa_H - \kappa_L) + \frac{b}{s} \kappa_L \right] m_H^2.$$

The second derivative with respect to m_H is:

$$\frac{d^2 \Pi_{\kappa_H}}{dm_H^2} = - \left[\mu_A (\kappa_H - \kappa_L) + \frac{b}{s} \kappa_L \right] = - \frac{\mu_A s (\kappa_H - \kappa_L) + b \kappa_L}{s} < 0,$$

which is strictly negative everywhere on $[0, k]$ because $\kappa_H > \kappa_L$, $\mu_A > 0$, $s > 0$, and $b > 0$. Therefore, along the binding IC_L boundary, the leader's objective is globally strictly concave in m_H , with unique interior maximizer:

$$m_H^* = \frac{b \Lambda}{\mu_A s (\kappa_H - \kappa_L) + b \kappa_L} = \frac{\Lambda \tilde{\rho}}{s (\kappa_H - \kappa_L) + \tilde{\rho} \kappa_L},$$

where $\tilde{\rho} \equiv b/\mu_A = (\lambda_A - c_Q)/\mu_A$.

Evaluating the asking rate at this global boundary optimum:

$$a_H^{SB} = a_H(m_H^*) = 1 - \frac{U(m_H^*, 1; \kappa_L) - U_L^0}{s(k - m_H^*)}.$$

Because $s(k - m_H^*) > 0$ for all $m_H^* < k$, strict downward distortion $a_H^{SB} < 1$ holds if and only if $U(m_H^*, 1; \kappa_L) - U_L^0 > 0$. Expanding this difference using $U(m_H^*, 1; \kappa_L) = V - c_Q(k - m_H^*) - \frac{\kappa_L}{2} (m_H^*)^2$:

- When type L 's unconstrained effort is interior ($k \geq \Lambda/\kappa_L$), $U_L^0 = V - \Lambda k + \frac{\Lambda^2}{2\kappa_L}$. The rent difference becomes:

$$U(m_H^*, 1; \kappa_L) - U_L^0 = sk - \frac{\Lambda^2}{2\kappa_L} + c_Q m_H^* - \frac{\kappa_L}{2} (m_H^*)^2 = s \left[k - \frac{\Lambda + c_Q}{2\kappa_L} \right] - \frac{(c_Q - \kappa_L m_H^*)^2}{2\kappa_L}.$$

Setting this difference strictly positive yields the explicit parameter inequality:

$$k > k_{\text{crit}}^{\text{int}} \equiv \frac{\Lambda + c_Q}{2\kappa_L} + \frac{(c_Q - \kappa_L m_H^*)^2}{2s\kappa_L}.$$

- When type L 's unconstrained effort saturates at the boundary ($k < \Lambda/\kappa_L$), $U_L^0 = V - \frac{\kappa_L}{2} k^2$. The difference factors as:

$$U(m_H^*, 1; \kappa_L) - U_L^0 = (k - m_H^*) \left[\frac{\kappa_L(k + m_H^*)}{2} - c_Q \right],$$

which is strictly positive if and only if:

$$k > k_{\text{crit}}^{\text{clip}} \equiv \frac{2c_Q}{\kappa_L} - m_H^*.$$

Combining both cases defines the threshold:

$$k_{\text{crit}}(\kappa_L, \kappa_H, \Lambda, c_Q, \lambda_A, \mu_A) \equiv \begin{cases} \frac{\Lambda + c_Q}{2\kappa_L} + \frac{(c_Q - \kappa_L m_H^*)^2}{2s\kappa_L}, & \text{if } k \geq \frac{\Lambda}{\kappa_L}, \\ \frac{2c_Q}{\kappa_L} - m_H^*, & \text{if } k < \frac{\Lambda}{\kappa_L}. \end{cases} \quad (1)$$

Whenever $k > k_{\text{crit}}$, $U(m_H^*, 1; \kappa_L) > U_L^0$ holds strictly, which guarantees:

$$a_H^{SB} < a_H^B \leq a_H^{FB} = 1.$$

In our representative calibration ($k = 10, \Lambda = 5, c_Q = 2, s = 3, b = 2, \mu_A = 1, \kappa_L = 0.3, \kappa_H = 0.5$), $\Lambda/\kappa_L = 50/3 > k = 10$, so the clipped branch applies, giving $m_H^* = 25/3 \approx 8.33$ and $k_{\text{crit}} = \frac{4}{0.3} - \frac{25}{3} = 5.0 < k = 10$. The condition is satisfied with a 100% margin, producing $a_H^{SB} = 0.75 < 1$ exactly matching the numerical second-best contract.

We now analytically demonstrate that maintaining $a_H = 1$ is globally strictly suboptimal. Let $\mathcal{F}_H = \{(m_H, a_H) \in [0, k] \times [0, 1] : U(m_H, a_H; \kappa_L) \leq U_L^0\}$ denote the full feasible set of high-cost contracts satisfying IC_L . Because $\partial U / \partial a_H = s(k - m_H) > 0$ for all $m_H < k$, any feasible bundle $(m_H, a_H) \in \mathcal{F}_H$ must satisfy:

$$a_H \leq a_H(m_H) \equiv 1 - \frac{U(m_H, 1; \kappa_L) - U_L^0}{s(k - m_H)}.$$

For any fixed m_H , the leader's payoff $\Pi_{\kappa_H}(m_H, a_H) = \mu_A U(m_H, a_H; \kappa_H) - b a_H (k - m_H)$ has partial derivative:

$$\frac{\partial \Pi_{\kappa_H}}{\partial a_H} = (\mu_A s - b)(k - m_H).$$

If $\mu_A s - b \leq 0$, Π_{κ_H} is non-increasing in a_H , so choosing $a_H = 1$ is immediately weakly or strictly dominated by choosing a smaller asking rate. If $\mu_A s - b > 0$, Π_{κ_H} is strictly increasing in a_H . For any fixed m_H , the leader maximizes surplus within $\mathcal{F}_H$ by setting a_H at its upper bound:

$a_H = \min\{1, a_H(m_H)\}$. Along the locus where this bound binds at $a_H = a_H(m_H)$, the leader's reduced objective $\Pi_{\kappa_H}(m_H)$ is strictly concave in m_H on $[0, k]$ with unique global maximizer m_H^* . Because $k > k_{\text{crit}}$, we have $a_H(m_H^*) < 1$. Now consider any candidate contract $(\tilde{m}_H, 1) \in \mathcal{F}_H$ that maintains $a_H = 1$. Feasibility under IC_L requires $U(\tilde{m}_H, 1; \kappa_L) \leq U_L^0$, which is equivalent to $a_H(\tilde{m}_H) \geq 1$. Because $a_H(m_H^*) < 1$, any such candidate must have $\tilde{m}_H \neq m_H^*$. By the strict concavity of $\Pi_{\kappa_H}(m_H)$, it follows that:

$$\Pi_{\kappa_H}(\tilde{m}_H, 1) \leq \Pi_{\kappa_H}(\tilde{m}_H, a_H(\tilde{m}_H)) < \Pi_{\kappa_H}(m_H^*, a_H(m_H^*)) = \Pi_{\kappa_H}(m_H^{SB}, a_H^{SB}).$$

In our representative calibration, the best contract on the boundary $a_H = 1$ requires $\tilde{m}_H = k = 10$, yielding leader payoff $\Pi_{\kappa_H}(10, 1) = 75.0$, whereas the interior contract $(8.33, 0.75)$ delivers $\Pi_{\kappa_H}^{SB} = 75.5556$, an exact surplus gap of $5/9 > 0$. Hence, maintaining $a_H = 1$ is globally strictly suboptimal, and the high-cost type's asking service is strictly distorted downward: $a_H^{SB} < a_H^B \leq 1$.

Step 5: Co-activity of IC_H under extreme bias. As bias λ_A increases, maintaining IC_L forces a_H^{SB} progressively lower. When a_H^{SB} approaches $a_L^{SB} = 0$, the high-cost type receives negligible clarification assistance relative to its higher specification cost. Eventually, the reverse incentive constraint IC_H ($U(m_H, a_H; \kappa_H) \geq U(m_L, a_L; \kappa_H)$) can become co-active. Under sufficiently extreme bias, separating the types via distinct asking rates is no longer incentive-viable, and the optimal screening menu may collapse to a pooling or boundary-saturated allocation where both types receive identical asking service. $\square$

Remark 7 (Active constraint structure without monetary transfers). *In classical transfer-based screening (e.g., Baron–Myerson), IC_L and IR_H typically bind while IC_H and IR_L remain slack. In our setting, solving the full four-variable constrained problem shows that IC_L binds alone, with IR_H , IC_H , and IR_L remaining strictly slack at representative under-asking-bias parameters. This structural divergence arises because the leader's payoff Π_{κ_H} directly internalizes follower welfare $\mu_A U(\cdot; \kappa_H)$, eliminating the classical principal's motive to compress agent rent down to the reservation utility. We note that the exact active constraint configuration is parameter-region-specific: under sufficiently extreme asking friction ($\lambda_A \gg c_Q$) or severe discounting ($\mu_A \ll 1$), IC_H can bind alongside IC_L . The robust qualitative insight is that screening under misalignment strictly distorts $a_H^{SB} < a_H^B$, while the active constraint configuration reflects the non-monetary, partially aligned nature of the objective.*

Remark 8. *The direction in Proposition 6 is specific to the under-asking bias case ($\lambda_A > \mu_{ACQ}$), which we take to be the empirically relevant one (Section 3.3's motivating story: turn-level satisfaction signals penalize extra clarifying questions more than users actually mind answering them). Under the opposite bias, the direction of distortion can flip; we do not claim a sign that holds for all Π_{κ} .*

7 Comparing the Two Regimes

Corollary 6 (Menus weakly dominate pooling). *The leader's optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy a is a feasible (degenerate) menu with $a_L = a_H = a$, trivially satisfying IC , so the unconstrained menu optimum is at least as good.*

Beyond payoff dominance, the central comparison lies in the structural decomposition of welfare losses across the two regimes, enabled by Propositions 4–6:

- **Model I (pooling)** incurs welfare loss from two distinct sources: (i) the pooling restriction itself, which operates even when $\mu_A=1$ and $\lambda_A=c_Q$, where it manifests as a corner solution that fully serves one type while completely neglecting the other (Corollary 5); and (ii) objective bias, which shifts a^{SE} (Corollary 2) and determines whether the equilibrium settles at a corner or an interior rate. These two forces interact non-linearly across the parameter space.
- **Model II (screening)** incurs welfare loss from bias *alone* (Proposition 5): in the absence of bias, screening recovers full efficiency despite private cost information. Under bias, the loss takes the form of the downward distortion characterized in Proposition 6. This represents a fundamentally different mechanism from the pooling loss: it stems not from an inability to condition on observed effort, but from the requirement to prevent opportunistic mimicking across types.

7.1 Sensitivity of the Unbiased Pooling Corner and Physical Feasibility

While the qualitative finding of Corollary 5, that unbiased pooling is always a corner solution rather than an interior compromise, holds universally across the parameter space, the unclipped corner difference formula $\Pi(1) - \Pi(0) = (\Lambda - c_Q)[k - k^*]$ relies on unconstrained best responses. Table 1 documents this unconstrained boundary across varying task sizes k and baseline guessing accuracies g , holding $L = 10, c_Q = 2$, and $\kappa \sim \text{Uniform}[0.2, 0.6]$ (where $\mathbb{E}[1/\kappa] \approx 2.747$).

g	$\Lambda = L(1 - g)$	Threshold k^*	Task Size k	Unclipped $\Delta\Pi$	Clipped $\Delta\Pi$
0.30	7.0	12.36	5	-36.80	+0.15
0.30	7.0	12.36	10	-11.80	+1.48
0.30	7.0	12.36	15	+13.20	+13.20
0.50	5.0	9.61	5	-13.84	+0.50
0.50	5.0	9.61	9	-1.84	+0.32
0.50	5.0	9.61	10 (Benchmark)	+1.16	+1.16
0.50	5.0	9.61	12	+7.16	+7.16
0.50	5.0	9.61	15	+16.16	+16.16
0.70	3.0	6.87	5	-1.87	+0.36
0.70	3.0	6.87	8	+1.13	+1.13
0.70	3.0	6.87	10	+3.13	+3.13

Table 1: Sensitivity of the unbiased pooling corner choice across task complexity k and guessing capability g , for $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$, where $\Delta\Pi \equiv \Pi(1) - \Pi(0)$. Columns compare the unconstrained closed form $(\Lambda - c_Q)[k - k^*]$ against the physically feasible clipped payoff difference. In the unclipped algebra, guessing appears to win whenever $k < k^*$ (e.g., -36.80 at $g = 0.30, k = 5$), but when followers are constrained to feasible effort $m \leq k$, asking strictly dominates across all rows (+0.15 at $g = 0.30, k = 5$), confirming Proposition 3.

As shown in Table 1, once followers are restricted to feasible specification effort $m \in [0, k]$, asking ($a^{SE} = 1$) weakly dominates guessing across all 11 configurations under linear risk. The apparent dominance of guessing in the unclipped column stems entirely from low-cost types choosing unfeasible specification levels $m = \Lambda/\kappa > k$. In contrast, when evaluated under the exact conjunctive success function $q(a, g)^{k-m}$, non-linear probability curvature generates genuine interior trade-offs where guessing beats asking even on the interior domain (Section 10).

Remark 9 (Compounding, testable). *Because Model II's distortion is triggered by, and scales with, the same bias parameters (μ_A, λ_A vs. c_Q) that drive Model I's shift in a^{SE} , reducing measured*

assistant bias should improve outcomes for high-cost (terse) users under both regimes, but the welfare gap between menu-conditioned and pooled policies should track population heterogeneity in κ (the spread $\kappa_H - \kappa_L$ and how far F sits on either side of κ^) almost independently of bias. This gives two separable, testable quantities from the same field data: a bias-sensitive term and a heterogeneity-sensitive term.*

8 Testable Predictions

- P1** (Prop. 1, Cor. 1) Holding the assistant’s policy fixed, higher guessing accuracy is associated with shorter specifications alongside higher or equal success rates, and never with shorter specifications paired with lower success rates.
- P1’** (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length flips sign depending on whether $c_Q \geq \Lambda = L(1 - g)$: where clarifying questions are inexpensive relative to the risk they resolve, users specify less as the ask rate rises (substitution); where questions are relatively costly to answer, users specify *more* (pre-emption). This prediction operates within the model and is distinct from P1’s accuracy channel.
- P2** (Prop. 2) The correlation between turn-level satisfaction and eventual task success weakens or inverts specifically for under-asking assistants, and can be benchmarked against the derived a^{SE} of Proposition 4 rather than an assumed ask rate.
- P3** Making ask-rate statistics visible to users shifts specification length toward the sophisticated best response analyzed in Section 3.3.
- P4** (Prop. 3) Ask rates should rise with a user’s revealed cost of providing detail (terseness) when clarification is constrained or heterogeneous, rather than responding exclusively to model confidence.
- P5** Users with lower self-reported error-catching rates exhibit declining specification effort across an interactive session (following Section 3’s dynamic channel).
- P6** (Cor. 5) Under a pooling policy, terse users are under-served and verbose users are over-served relative to *each type’s own* first best, even for a fully unbiased assistant. This efficiency gap is attributable purely to instrument restriction; it can be distinguished empirically from bias by holding λ_A fixed (for instance, via controlled variation in training friction penalties) and varying only whether the policy can condition on declared specification effort.
- P7** (Prop. 6) Even when an assistant *can* condition on declared specification effort, terse users receive fewer clarifying questions than the assistant’s own per-type preference would prescribe. This distortion arises specifically from the incentive-compatibility constraint and contracts as population cost heterogeneity ($\kappa_H - \kappa_L$) diminishes, holding bias fixed.

9 Empirical Strategy and Experimental Design

The behavioral predictions P1–P7 translate our contract-theoretic mechanisms into observable margins for experimental testing. In naturalistic software engineering logs, however, assistant policy bias λ_A and user specification costs κ are typically confounded with task difficulty and user experience. To causally test our predictions, we propose a between-subjects randomized controlled trial on software developers performing standardized implementation tasks:

Experimental design and treatments. Participants are randomly assigned to interact with code assistants operating under distinct clarification regimes:

1. *Treatment A (Low-asking policy, $a = 0.1$):* The assistant aggressively guesses ambiguous attributes with minimal clarification.
2. *Treatment B (High-asking policy, $a = 0.8$):* The assistant frequently asks targeted clarifying questions before generating code.
3. *Treatment C (Menu-conditioned screening policy):* The assistant dynamically selects its clarification rate based on the structural completeness of the initial user prompt, implementing the screening menu derived in Section 6.
4. *Control (Unbiased benchmark):* The assistant’s clarification rate is calibrated to the user-optimal benchmark.

Measurement and identification. For each programming session, instrumentation logs: (i) upstream prompt length and specified invariants/edge cases (m); (ii) clarification turn count ($a(k - m)$); (iii) downstream test suite execution and correctness; and (iv) post-session subjective developer satisfaction. By exogenously varying the clarification policy a , this protocol directly tests whether users economize on prompt specification in response to anticipated assistance, distinguishes the substitution channel ($\Lambda > c_Q$) from the pre-emption channel ($c_Q > \Lambda$, Prediction P1’), and isolates the welfare losses attributable to objective bias versus pooling restrictions (Predictions P6 and P7).

10 Discussion and Limitations

The linear-risk approximation to the conjunctive success function is the main tractability assumption carried through Sections 4–7. To verify that this simplification does not drive our core theoretical findings, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a, g)^{k-m}$ (independently of the linear-risk closed forms, via direct numerical grid search and constrained optimization). Across wide ranges of task complexity ($k \in [3, 20]$) and guessing accuracy ($g \in [0.50, 0.95]$), numerical optimization confirms that the qualitative structure survives under the exact nonlinear specification: the corner-solution property of Corollary 5 persists ($a_{\text{exact}}^{SE} \in \{0, 1\}$); downward distortion ($a_H^{SB} < a_H^B$) persists robustly, with exceptions concentrated at small k where specification effort saturates at $m = k$; and participation constraints remain strictly slack. Furthermore, numerical solutions of the full four-variable constrained menu problem under the exact conjunctive payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ confirm that IC_H binds alongside IC_L under extreme bias. This reproduces the three-regime structure (no binding constraint, IC_L alone, and $IC_L + IC_H$ together) identified under the linear-risk approximation, with only the boundary between moderate- and extreme-bias regimes shifting slightly (from $\lambda_A \approx 6.0$ to $\lambda_A \approx 8.0$ at $\mu_A = 1.0$). The extreme-bias reversal is therefore an intrinsic structural feature of the screening contract space, rather than an artifact of the linear-risk approximation.

Regarding the regularity condition (Assumption 1), monotonicity of prompt specification ($m^*(g)$ weakly decreasing in g) holds whenever $(k - m) \ln q \geq -1$. While monotonicity can fail when the cross-partial turns strictly positive, empirical monotonicity can still survive locally in low-stakes regimes as a boundary property, because Topkis submodularity is a sufficient rather than necessary condition for comparative statics.

A stylized modeling assumption is attribute symmetry (k binary attributes with identical loss L and guessing probability g). In practical software engineering, attributes exhibit substantial loss heterogeneity: omitting an architectural invariant or security convention causes severe failure ($L_i \gg 0$), while omitting a stylistic naming convention carries negligible cost ($L_j \approx 0$). In an extension with attribute-specific losses L_i , the user’s objective becomes $U = V \prod_i q_i - c(m) - \sum_i a_i c_Q$. Ordering attributes by descending risk stake $\Lambda_i = L_i(1 - g_i)$, users rationally specify the highest-stakes attributes first, while assistants prioritize asking on the highest-residual-stake attributes. The fundamental contract-theoretic mechanism, namely the substitution between upstream specification effort and automated clarification under misaligned asking friction, generalizes directly to the heterogeneous-loss setting.

A related limitation concerns the empirical identification of λ_A . Every bias-driven prediction (Corollary 2, Proposition 6, and predictions P2, P6, P7) is stated in terms of $\lambda_A > \mu_{ACQ}$, but backing out λ_A from observed policy alone is circular. Corollary 3 formalizes this as an exact non-identification result on the interior branch: a^{SE} depends on (μ_A, λ_A) only through their ratio, so a discounting-driven leader and a friction-driven leader with the identical ratio λ_A/μ_A cannot be distinguished from pooling behavior alone. An independent empirical handle, such as an explicit penalty on turn count in the training objective or reward specification, would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested without circularity. The two-type restriction in Model II avoids ironing complications at the cost of generality; a continuum formulation using the standard Mirrlees technique is a natural next step that would allow κ^* from the first-best benchmark to interact directly with the screening distortion. We also assume truthful revelation of specified attributes (abstracting from cheap-talk ambiguity within what is communicated) and a single leader. Modeling competition among multiple assistants represents an appealing extension, as market competition could plausibly discipline asking bias directly.

A broader methodological conclusion concerns the limitations of applying classical contract-theoretic intuition directly to human–AI settings without accounting for non-monetary, partially aligned incentives. In classical mechanism design (such as Baron–Myerson or Laffont–Tirole), the principal pays monetary transfers to extract surplus, and agent informational rent directly reduces the principal’s payoff. In human–AI interaction, the assistant’s objective directly incorporates user welfare. As a consequence, incentive compatibility is costless in the absence of bias (Proposition 5), participation constraints do not bind at the bottom of the type distribution, and unbiased pooling naturally induces corner solutions rather than interior compromises (Corollary 5). Economic analyses of automated assistance must therefore explicitly incorporate the non-monetary, partially aligned nature of the interaction rather than relying on direct analogies to cash-transfer mechanisms.

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